Digital Payments and Transaction Coordination Platform

Course : FFSD

Domain : Fintech

Team : NexusPay(46)

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PREFACE

In recent years, digital payment technologies have become an integral part of modern financial systems, enabling secure, fast, and efficient transactions across various platforms. The increasing demand for reliable digital transaction management has highlighted the need for well-designed systems that can coordinate payments, ensure data accuracy, and maintain transaction integrity. This Full-Stack Software Development (FFSD) project, titled **“Digital Payments and Transaction Coordination Platform,”** is developed to address these requirements.

This project focuses on designing and implementing a platform that facilitates digital payment processing while ensuring seamless coordination between transactions, users, and system components. The application aims to provide a structured and secure environment for handling payment-related operations, emphasizing usability, scalability, and performance.

The purpose of this document is to present a clear and comprehensive overview of the project, its objectives, and its scope. It serves as a reference for understanding the system’s functionality and the development approach followed during the project lifecycle. Through this project, practical knowledge of full-stack development concepts is applied to build a real-world fintech-oriented solution.

INTRODUCTION

The advancement of digital technologies has significantly transformed the financial sector, leading to the widespread adoption of digital payment systems. Digital payments enable users to perform financial transactions electronically, reducing dependency on cash and improving transaction speed, convenience, and transparency. As the volume of online transactions continues to grow, the need for efficient coordination and management of these transactions has become increasingly important.

The **Digital Payments and Transaction Coordination Platform** is designed to provide a centralized system for managing digital payment operations and ensuring accurate coordination between transactions, users, and system services. The platform focuses on securely processing payments, tracking transaction statuses, and maintaining consistency across multiple components involved in a transaction workflow.

This system aims to address common challenges such as transaction failures, data inconsistency, delayed confirmations, and security concerns. By implementing structured transaction coordination mechanisms and secure communication between system components, the platform ensures reliability and integrity of financial data. The project applies full-stack development principles to integrate frontend interfaces, backend services, and databases into a cohesive and scalable solution suitable for modern digital payment environments.

User Requirements

Customer:

- Make digital payments using supported payment methods.
- Split Payment
- Scheduled Payment
- View transaction history with details such as date, amount, and Payee.

Merchant:

- Receive payments from customers.
- View incoming transactions and their status.
- Access transaction history and settlement details.
- Generate Annual transaction summaries or reports.

NexusPay Admin:

- Monitor all transactions across the platform.
- Handle transaction failures and coordination issues.
- Respond to user complaints and resolve disputes.
- Manage system configurations and permissions.

Bank Admin:

- Manage bank-side transaction processing.
- View transaction logs related to the bank.
- Handle settlement and reconciliation processes.

System Requirements

Functional Requirements:

1. Authentication & Role Management:

- The system shall authenticate participants using secure credentials.
- The system shall authorize actions based on assigned role authority (payer, admin, merchant).
- The system shall allow profile creation and maintenance.

2. Digital Payment Execution:

- The system shall allow a payer to initiate peer-to-peer payments.
- The system shall support bank-to-bank, bank-to-wallet, wallet-to-bank transfers.

3. Scheduled Payments:

- The system shall allow payers to create scheduled payments to a payee.
- The system shall automatically trigger payments on predefined dates.
- The system shall allow modification or cancellation of scheduled payments.

4. Transaction State Management:

- The system shall track transaction states throughout the payment lifecycle.
- The system shall update transaction state as initiated, authorized, pending, successful, or failed.
- The system shall store immutable transaction records.

5. Transaction Limits & Balance Validation:

- The system shall enforce spending limits per payer.
- The system shall validate funding source balance before executing payments.

6. Split Payments & Expense Groups:

- The system shall allow creation of expense groups.
- The system shall calculate expense splits among group participants.
- The system shall track individual contributions and settlement status.

7. Notifications & Alerts:

- The system shall notify payers and payees of payment events.
- The system shall generate alerts for failed or scheduled transactions.

8. Dispute & Refund Management:

- The system shall allow payers to raise dispute cases for transactions.
- The system shall allow NexusPay Admin to review dispute cases.
- The system shall process refund instructions for approved disputes.

9. Spending Analytics & Categorization:

- The system shall categorize transactions into spending categories.
- The system shall generate spending insights based on transaction history.
- The system shall provide summary reports for financial analysis.

10. Funding Source & Beneficiary Management:

- The system shall allow linking and management of funding sources.
- The system shall allow creation and verification of beneficiary profiles.
- The system shall restrict payments to verified beneficiaries only.

Non-Functional Requirements:

1. Security:

- The system shall prevent unauthorized access to financial actions.
- The system securely records all transaction details for future reference.

2. Performance:

- Payment validation shall be completed within an acceptable response time.
- Scheduled payments shall be triggered within the defined execution window.
- Notification delivery shall be near real-time.

3. Scalability:

- The system shall support an increasing number of payment transactions.
- The system shall handle concurrent scheduled payments efficiently.

4. Data Integrity & Consistency:

- Transaction records shall be immutable once created.
- Payment execution shall ensure atomic debit-credit operations.
- No duplicate or inconsistent transaction entries shall exist.

5. Auditability & Compliance:

- All financial actions shall be logged with timestamps.
- Dispute resolutions shall be traceable.
- Reports shall be available for administrative review.

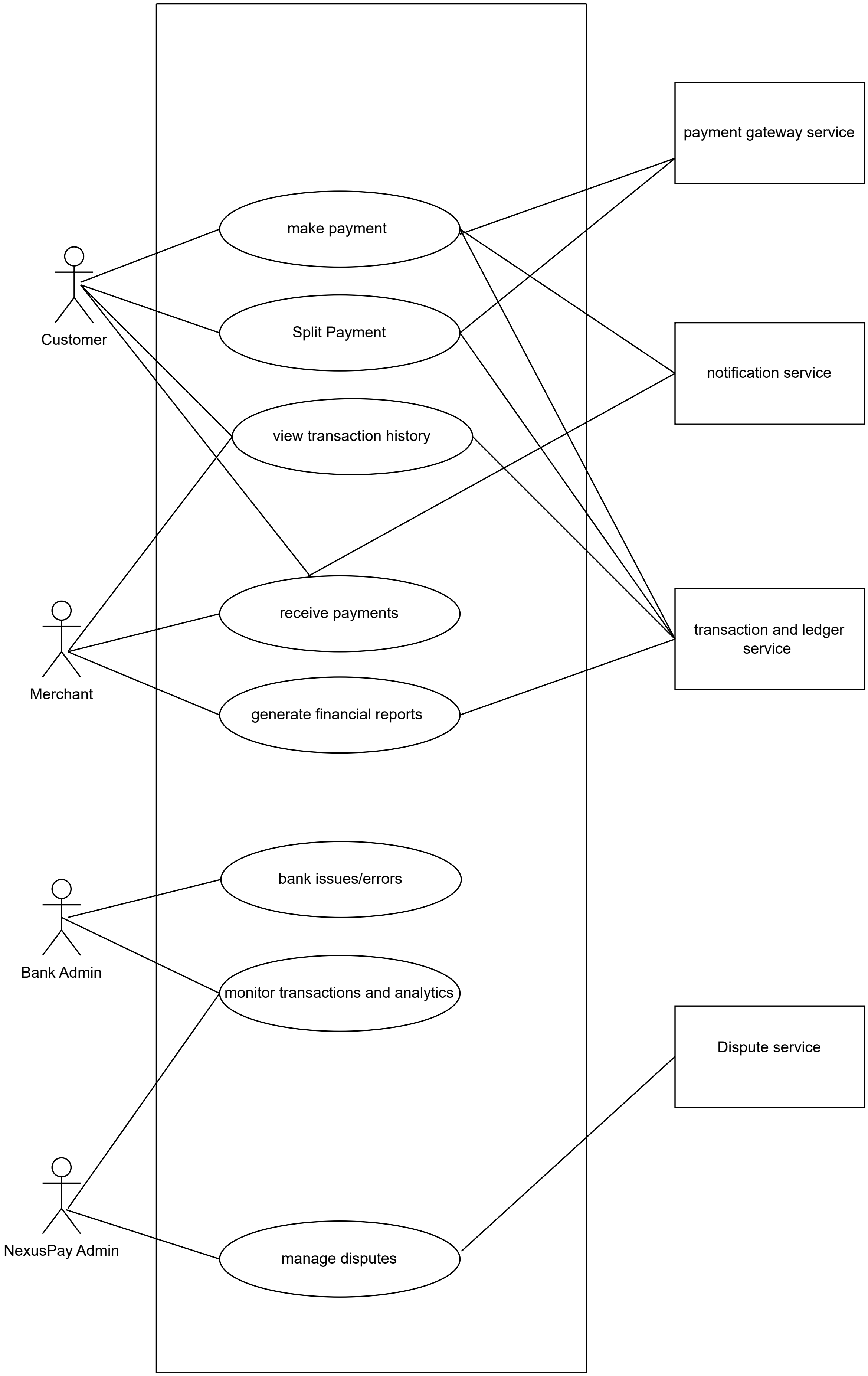
6. Maintainability:

- Domain terms shall be used consistently across code and documentation.
- Changes to business rules shall be easy to implement.

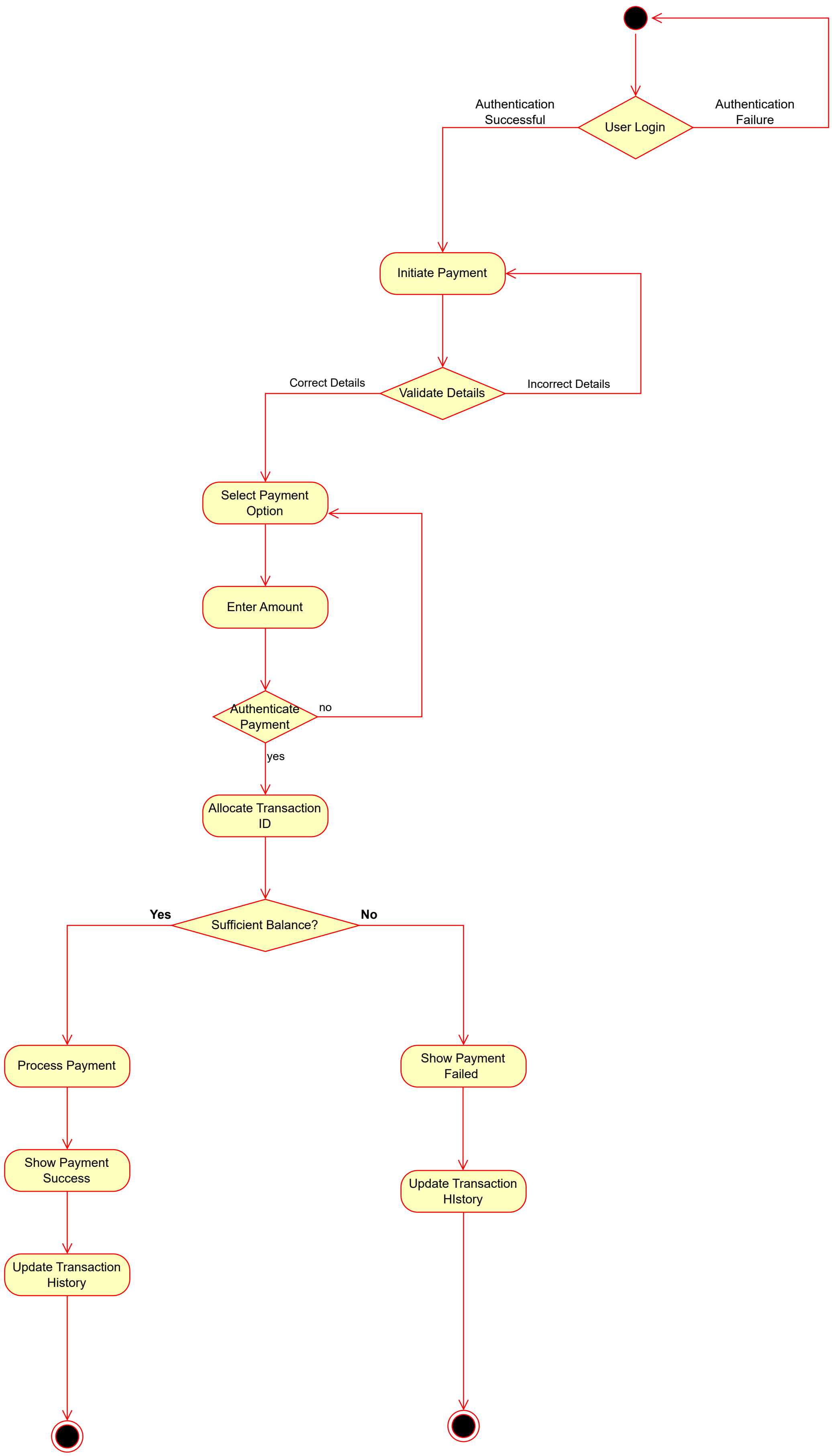
7. Compatibility & Interoperability:

- The system shall support integration with external banking systems.
- The system shall be compatible across modern browsers and devices.

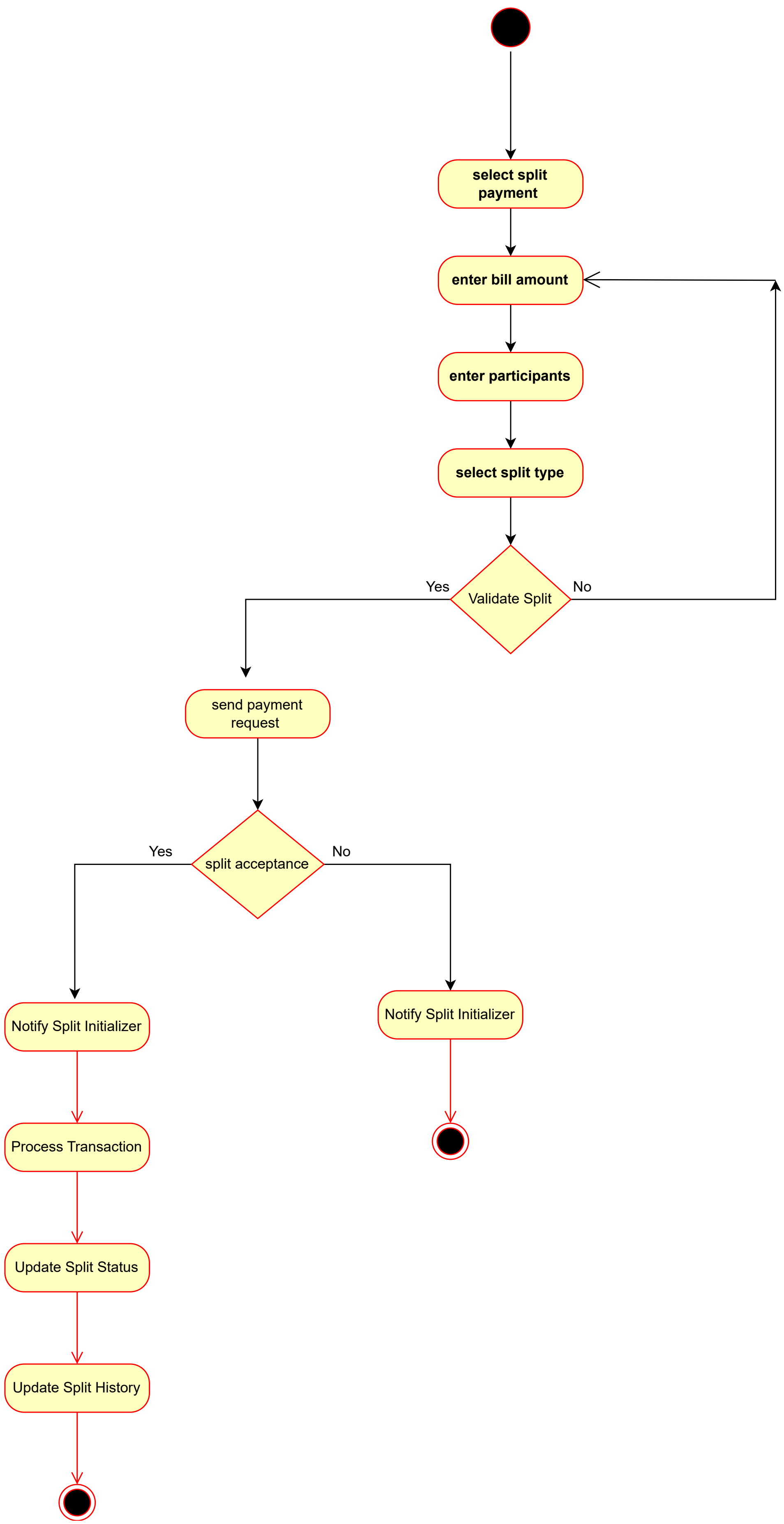
Use - Case Diagram



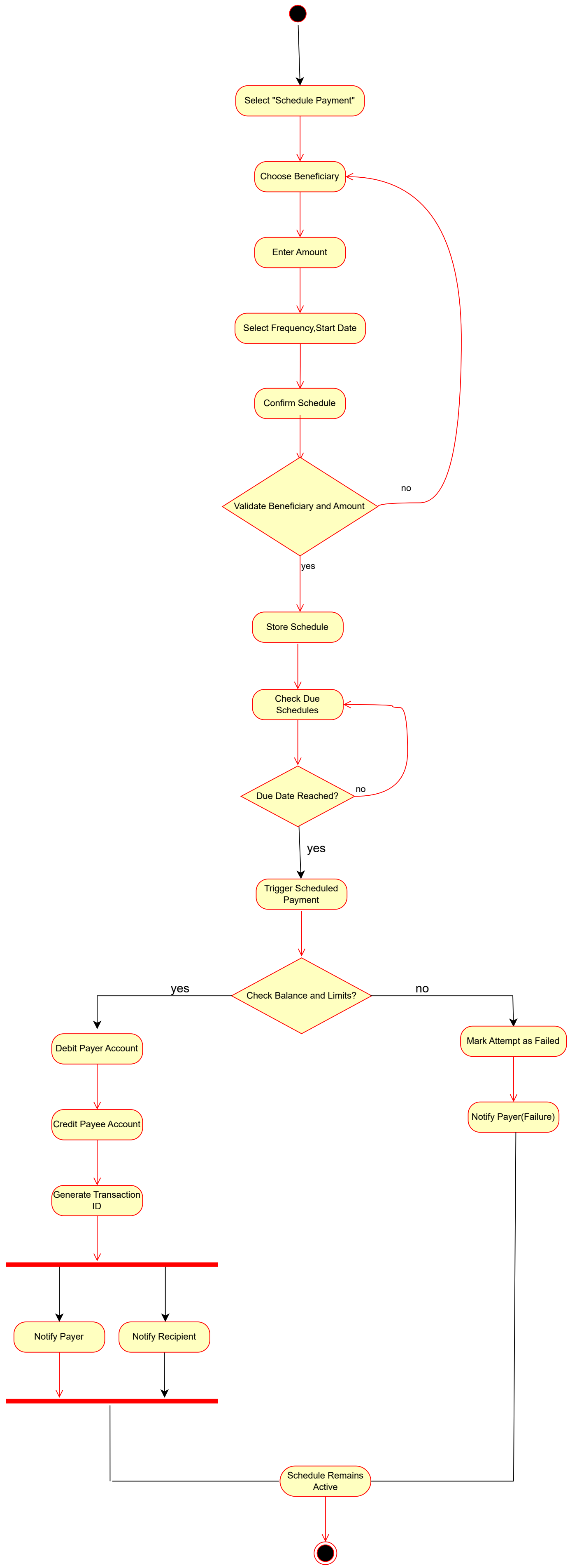
Make Payment Activity Diagram



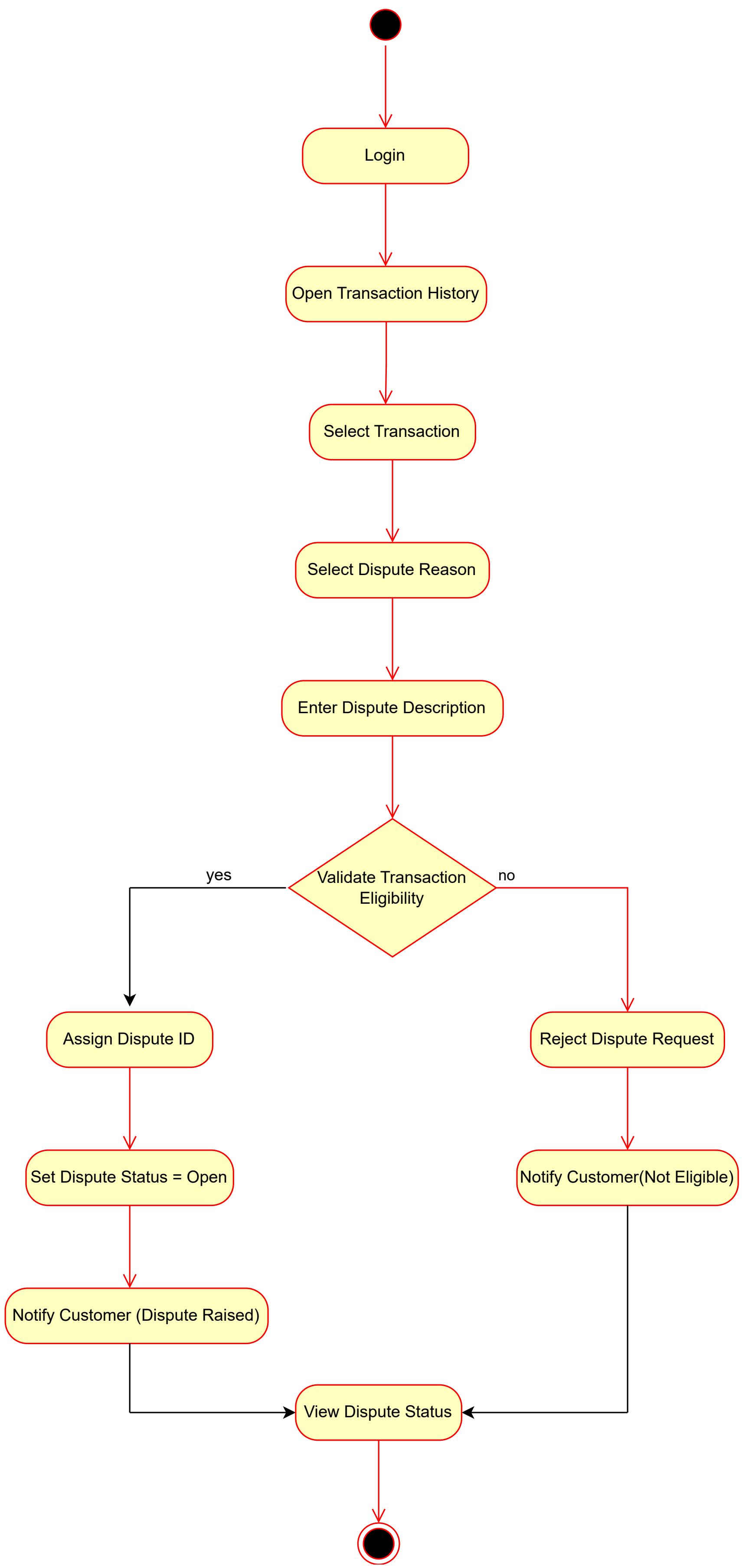
Split Activity Diagram



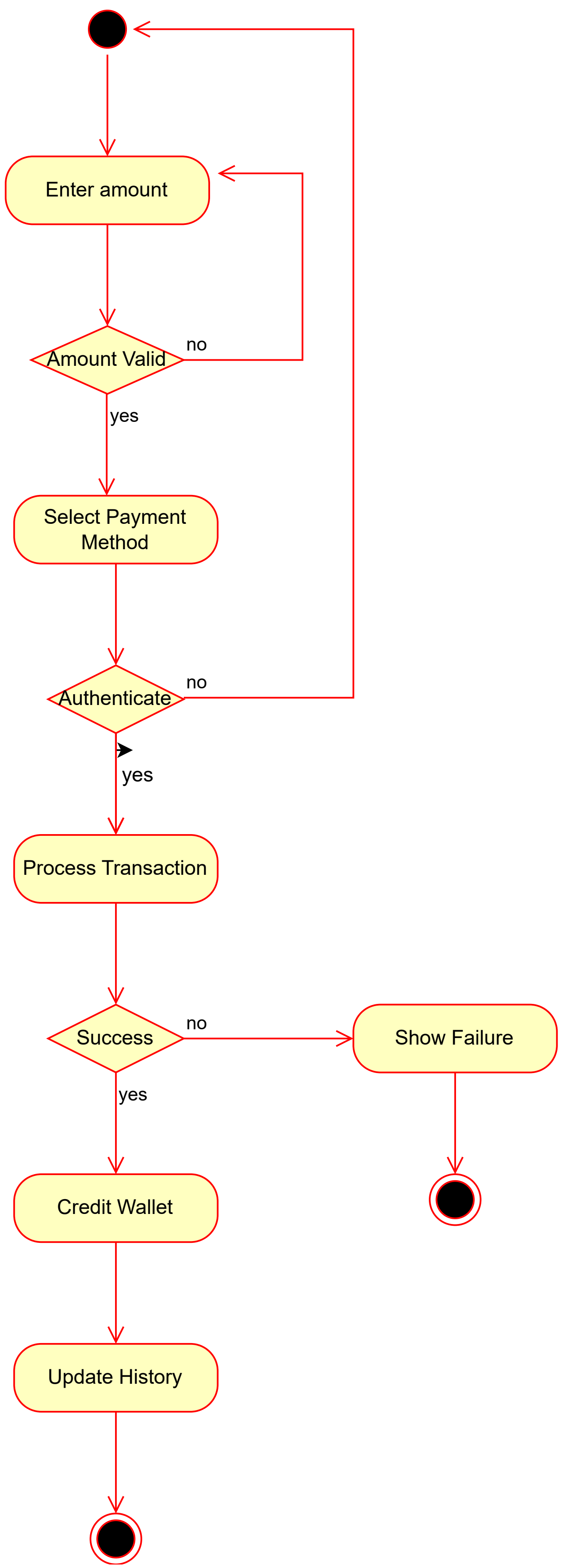
Schedule Payment Activity Diagram



Raise Dispute Activity Diagram



Bank To Wallet Payment



NexusPay - Consolidated Payment, Split, Scheduled & Dispute Sequence